



BURKINA FASO

COUNTRY REPORT ON GENOME EDITING LANDSCAPE

ANALYSIS ON AGRICULTURE

MAY 2026

TABLE OF CONTENTS

LIST OF TABLES.....	iii
LIST OF FIGURES.....	iii
LIST OF ABBREVIATIONS AND ACRONYMS.....	iv
EXECUTIVE SUMMARY.....	vi
1. INTRODUCTION.....	1
1.1. Description of Burkina Faso's Agricultural Background.....	1
1.2. National Frameworks and Investments.....	1
1.3. Engagement in Regional Frameworks.....	2
1.4. Challenges and Opportunities: Justification of GEd Assessment based on Current Issues Identified in Burkina Faso.....	2
1.4.1. Challenges Justifying GEd Assessment.....	2
1.4.2. Opportunities Supporting GEd Investment.....	3
1.5. Objectives of the Landscape Analysis	3
2. APPROACH/METHODOLOGY	5
2.1. Status of Policy and Regulatory Frameworks Relevant to Emerging Biotechnologies, Including Genome Editing	5
2.2. Projects, Crops, Livestock, Fisheries, Forestry and Traits Ready for Commercialisation And Scaling	5
2.3. Staple, Traditional and Commercial Crops Important to Burkina Faso that Could be Improved Using GEd Technology	5
2.4. Institutional Capacity (Human Capital, Laboratory and Field Infrastructure, Equipment).....	6
2.5. Stakeholder Mapping Relevant for Genome Editing Research-Development and Future Commercialisation (or Deployment)	6
2.6. Database Systems and Database Management.....	6
2.7. Data Synthesis and Data Analysis	7
2.8. Interactive Map.....	7
3. RESULTS	8
3.1. Membership in Multilateral Environment Treaties and Agreements.....	8
3.2. Policies that Advance or Hinder Adoption of Modern Biotech and GEd.....	9
3.3. National regulatory framework.....	9
3.3.1 Biosafety Act/Law, Biosafety Regulations/Decrees and GEd Guidelines	9

3.4. Socio-economic considerations for decision-making in GEEd technology and application	15
3.5. Factors that Hinder Application of Modern Biotech and GEEd	16
3.6. Genome Editing Programmes and Projects.....	17
3.7. Human Capital and Institutional Capacity	19
3.7.1 Women, Youth and Special Interest Groups' Involvement in Modern Biotech and GEEd	22
3.7.2. Training and Professional Development	22
3.8. Infrastructure and Equipment	24
3.8.1. Universities and Research Institutions.....	24
3.9. Traditional and Staple Crops, Livestock, Agroforestry, and Fisheries Varieties/ Breeds for Improvement Using GEEd	27
3.10. Intellectual Property Rights and Benefit Sharing.....	30
3.11. Private Sector Participation.....	30
3.12. Funding and Investment Landscape.....	32
4. RECOMMENDATIONS AND POLICY OPTIONS.....	36
4.1 Recommendations	36
4.2 Policy Options	37
REFERENCES	38
ANNEXES.....	39

LIST OF TABLES

Table 1: Status of Burkina Faso's participation in key multilateral environmental agreements	8
Table 2: Regulatory and institutional landscape for genome editing in Burkina Faso	12
Table 3: Genome editing projects and programmes in Burkina Faso.....	18
Table 4: Overview of academic and research institutions working on genome editing and related capacity in Burkina Faso.	20
Table 5: Overview of training programmes on genome editing	23
Table 6: Status and needs assessment of biosafety laboratory facilities by institution	26
Table 7: Priority organisms for genome editing application	28
Table 8: Burkina Faso's IPRs, legal instruments (laws) and organizations in charge	30
Table 9: Overview of private sector companies involved in seed production & distribution, crop improvement & biotech/GEd activities in Burkina Faso	32
Table 10: Overview of national and other funding sources for genome editing	33

LIST OF FIGURES

Figure 1: Decision making and notification process for GEd.....	14
Figure 2: CFT and release	14
Figure 3: Decision making process for genome editing	15

LIST OF ABBREVIATIONS AND ACRONYMS

ANB	Agence nationale de biosécurité
ANES-BF	Association Nationale des Entreprises Semencières du Burkina Faso (National Association of Seed Companies-Burkina Faso)
AUDA-NEPAD	African Union Development Agency – New Partnership for Africa’s Development
CBD	Convention on Biological Diversity
CEA/ITECH- MTV	Centre d’Excellence Africain / Innovation Technologique pour l’Élimination des Maladies à Transmission Vectorielle (African Center of Excellence in Biotechnological Innovations for the Elimination of Vector-Borne Diseases)
CERBA	Pietro Annigoni Biomolecular Research Centre
CIRDES	Centre International de Recherche-Développement sur l’Elevage en zone Subhumide
CNRST	Centre National de la Recherche Scientifique et Technologique
CNS	Comité National des Semences (National Seed Committee)
CRISPR	Clustered Regularly Interspaced Short Palindromic Repeats
DNA	Deoxyribonucleic Acid
ENS	Ecole Normale Supérieure
EPO	Ecole Polytechnique de Ouagadougou
ECOWAS	Economic Community of West African States
FAO	Food and Agriculture Organization of the United Nations
FENABIOTECH	Fédération Nationale des Agriculteurs en Biotechnologies
GDP	Gross Domestic Product
GE _d	Genome Editing
GMOs	Genetically Modified Organisms
GRAS	Health Action Research Group
HPV	Human Papillomavirus
IDR	Institut du Développement Rural (Institute of Rural Development)
INERA	Institut de l’Environnement et de Recherches Agricoles (Institute of Environment and Agricultural Research)
INSP	Institut National de Santé Publique (National Institute of Public Health)
IRD	Institut de Recherche pour le Développement
IRSS	Institut de Recherches en Sciences de la Santé (Institute of Research in Health Sciences)

LABIOGENE	Laboratoire de Biologie Moléculaire et de Génétique (Laboratory of Molecular Biology and Genetic Engineering)
LABIOTAN	Laboratoire de Biochimie, Biotechnologie, Technologie Alimentaire et Nutrition (Laboratory of biochemistry, biotechnology, food and nutritional technology)
MAAHMRAH	Ministre de l'agriculture, des aménagements hydro-agricoles, de la mécanisation, des ressources animales et halieutiques
MESRI	Ministère de l'Enseignement Supérieur, de la Recherche et de l'Innovation
NBA	National Biosafety Agency
NBO	National Biosafety Observatory
NBT	New Breeding Techniques
NGOs	Non-Governmental Organizations
NIPH	National Institute of Public Health
NSBC	Comité Scientifique National de Biosécurité (National Scientific Biosafety Committee)
OFAB	Open Forum on Agricultural Biotechnology
ONRIZ	Observatoire National du Riz (National Observatory of the Rice Sector in Burkina Faso)
RD-STIA	Organisation pour la Recherche et le Développement en Sciences, Technologies et Innovations Agricoles (Organization for Research and Development in Agricultural Science, Technology and Innovation)
PBR cowpea	Pod Borer Resistant Cowpea
SCHMFB	Sous-Comité d'Homologation des Matériels Forestiers de Base (SCHMFB) (Subcommittee for the release of Basic Forestry Materials)
SCHV	Sous-Comité d'Homologation des Variétés Agricoles (Subcommittee for the release of Agricultural Varieties)
SDGs	Sustainable Development Goals
STC	Scientific and Technical Council
UJKZ	Université Joseph KI-ZERBO (University Joseph KI-ZERBO)
UN	United Nations
UNB	University Nazi BONI
UNEP	United Nations Environmental Programme
UNFCCC	United Nations Framework Convention on Climate Change
UNPCB	Union Nationale des Producteurs de Coton du Burkina (National Union of Cotton Farmers of Burkina Faso)

EXECUTIVE SUMMARY

Background: Among African countries, Burkina Faso has been a pioneer in agricultural biotechnology, having commercialized Bt cotton and Pod Borer Resistant (PBR) cowpea in the pipeline for release. Genome editing (GE) technology is a powerful precision tool across sectors such as genetic engineering, agriculture, medicine, and environmental sustainability. Burkina Faso is currently experimenting with GE rice to address the country's critical agricultural challenges, including bacterial blight resistance and yield improvement.

The African Union Development Agency–New Partnership for Africa's Development (AUDA-NEPAD) commissioned a GE Landscape Analysis for Burkina Faso to assess the country's readiness to develop, deploy, and commercialize genome-edited products. The country's regulatory framework (policies, regulations, guidelines, and laws) and technical capacities (institutional, infrastructural, and human) for modern biotechnology and GE were evaluated to identify key enablers and constraints across scientific, socioeconomic, cultural, and governance dimensions. Opportunities for GE to address priority national needs at scale, including agricultural productivity, climate adaptation, and food and nutrition security, were reviewed. Additionally, the assessment also identified staple and traditional crops with high potential to improve livelihoods.

The assessment drew on secondary data (published literature and institutional databases) and primary data collected through interviews, surveys, and questionnaires. Findings were synthesized across areas including international treaty commitments, regulatory frameworks, ongoing GE projects and programmes, human and institutional capacity, research and training opportunities, priority crops and traits, intellectual property rights (IPRs), and funding mechanisms.

Key findings: Burkina Faso commits to global efforts in food safety, climate action, biodiversity conservation, and ozone layer protection, and to integrating these commitments into its food and environmental management frameworks. This is evidenced by its membership to a raft of relevant international legal instruments, including the Codex Alimentarius Commission (CAC), the Convention on Biological Diversity (CBD), the Cartagena Protocol, the Nagoya Protocol on Access and Benefit Sharing (ABS), the Nagoya-Kuala Lumpur Supplementary Protocol on Liability and Redress, the United Nations Framework Convention on Climate Change (UNFCCC), the Kyoto Protocol, and the Paris Agreement.

At the domestic level, Burkina Faso has a complete biosafety regulatory environment (Biosafety and environment-related laws, regulations, and guidelines) for research and development (R&D) and the commercialisation of modern biotechnology, including GEd products that have operated efficiently for decades. The Agence Nationale de Biosécurité (ANB), established by Decree No. 2004-262 of 2004, is the national biosafety competent authority within the Ministry of Higher Education, Research and Innovation, with mandate for biotechnology matters, including genome editing.

The Biosafety Act, No. 0064-2012, outlines the biotechnology approval process and identifies other regulatory agencies involved. Specialized guidelines for genome editing and other emerging technologies have also been developed to ensure that innovation is balanced with risk mitigation. A combination of institutions, such as the ANB, universities, and research institutes, provide a coordinated approach to advancing GEd. However, Burkina Faso still lacks a national policy or strategy that clearly articulates the government's official position on biotechnology and emerging technologies, including genome editing. Efforts are underway to address this gap.

Despite existing opportunities through public-private partnerships and regional collaborations, challenges remain, including limited investment in research, reliance on international funding, the absence of a national biotechnology policy and strategy, and intense anti-technology activism that has created high-level political ambivalence toward the technology. Notably, opponents of biotechnology do not distinguish between GEd and transgenics.

The Institute of Environment and Agricultural Research (INERA), in collaboration with the "Institut de Recherche pour le Développement" (IRD) in Montpellier, France, has been conducting a GEd research on rice for several years. Rice is a strategic food crop for food security in Burkina Faso, but its production is significantly constrained by bacterial blight caused by *Xanthomonas oryzae pv. Oryzae*. GEd technology is being used to develop bacterial-blight-resistant rice varieties to address this major constraint and unlock the potential of local rice production for food security.

The country's human capacity in GEd is primarily found in universities and national agricultural research organizations. Burkina Faso has nine public universities, two elite universities (Ecole Polytechnique of Ouagadougou, (Ecole Normale Supérieure), six university centres, and 17 private universities. Of these, two universities (University Joseph KI-ZERBO (UJKZ) and University Nazi BONI (UNB)) and seven research organizations (INERA, IRSS, CEA/ITECH-

MTV, INSP, CERBA, GRAS, CIRDES) have the capacity to train and conduct GEd-related research. Local scientists also have access to several GEd courses offered by international institutions, including the Innovative Genomic Institute, the Outreach Network for Gene Drive Research, Teaching and Research (Neuro) Science for Development (TReND), the Alliance for Science, and AUDA NEPAD.

In Burkina Faso, GEd socio-economic considerations focus on balancing food security, livelihoods, and market access. Key factors include the affordability of improved seeds or breeds, equitable access for smallholder farmers, and safeguards to avoid dependence on costly external inputs. It is widely believed that developing disease-resistant, higher-yielding crops could significantly enhance food security, reduce import costs, and empower smallholder farmers. Improved livestock and crop varieties could also open new markets and boost competitiveness in regional and global agricultural trade.

Socio-cultural acceptance, ethical debates, and public perception will play critical roles in shaping the trajectory of genome editing adoption in the country. Lessons from earlier experiences with GM crops, especially Bt cotton, indicate that capacity building, proactive science communication, farmer sensitization, and engagement with political leaders, policymakers, and civil society will be crucial to the successful deployment of emerging technologies such as genome editing.

Researchers in Burkina Faso are exploring how genome editing could improve traditional and commercial crops important to the country, while considering its application in health, livestock, and forestry. Some staple crops that could be improved through GEd include sorghum, millet, maize, rice, and cowpea, which are central to food security. Although important, these crops face challenges such as low yields, recurring drought, and pest infestations. Groundnuts, sesame, and cotton are also important cash crops. In agroforestry systems, shea and baobab trees provide pest resistance and nutritional value, while conserving the resilience of local landraces.

The Gates Foundation is the leading funder of GEd activities in Burkina Faso. The country also receives support from competitive sources, including the European Union, trusts, AUDA-NEPAD/African Biosafety Network of Expertise (ABNE), and the national government, which provide recurrent funding for in-country research and regulatory review. Other funding sources include GIZ, CGIAR institutions, and Open Philanthropy.

Policy implications

- **Short-term:** (12-24 months): communication and public engagement-managing the on-going misinformation, disinformation and mal-information surrounding GMOs is also affecting GEd.
- **Medium-term (2–5 years):** Include leveraging biotech for national development, and trade in biotech products while ensuring compliance with international standards. Fostering sustainable agricultural practices, improved food security, and promoting economic growth through advances in Biotech, especially GEd techniques and regional standing.
- **Long-term:** (5-10 years): Regional and international collaborations, sustained investment in genome editing R&D.

Conclusion: Burkina Faso has a complete regulatory framework, a committed scientific community, and a proven track record in agricultural biotechnology. The nation is well-positioned to harness genome editing for sustainable development. However, this potential can be realized only through government and private-sector funding, deliberate policy reform, investments in human capital, regulatory systems, and public engagement. Genome editing offers Burkina Faso an opportunity to improve agricultural productivity and achieve food and nutritional security; it will also be a pathway to improved health and sustainable growth in several other areas. Although genome editing offers specific advantages, it should be promoted as part of an integrated bio-innovation approach rather than in isolation.

1. INTRODUCTION

1.1. Description of Burkina Faso's Agricultural Background

In 2024, agriculture accounted for an estimated 18.59% of Burkina's GDP, up from 16.33% in 2023. The sector remains the backbone of employment, accounting for approximately 70% of the active population (World Bank, 2023). However, agriculture sector faces numerous challenges and constraints.

These challenges include climate variability, population pressure, and food insecurity. Abiotic constraints encompass the degradation of natural resources and irregular rainfall, while biotic constraints include damage caused by pests (insects, diseases, nematodes, weeds) and poor seed quality. GE_d marks a significant breakthrough in modern biotechnology, providing precise and efficient methods for genetic modification (Ansori et al., 2023). In Burkina Faso, this technology is being evaluated to address key challenges, particularly in agriculture, where food and nutritional insecurity, climate change, and pest problems remain prevalent (Tajudeen et al., 2022). Implementing New Breeding Techniques, such as genome editing, depends on a predictable, science-based regulatory framework, sound scientific research, and high-level political support.

The review of Burkina's GE_d landscape details the country's existing national legislative framework for GE_d and ongoing GE_d projects across research institutions and universities. The report also emphasizes the country's infrastructure, equipment, and human resources capacity. It also provides insight into the crops and traits that could be targeted for improvement through GE_d technology, whether for commercial or indigenous products. Additionally, the role of the private sector in GE_d development is addressed, and the review concludes with an overview of funding opportunities for GE_d research and development.

1.2. National Frameworks and Investments

The country needs a robust national framework to guide responsible development and sustainable application of biotechnology in general, and genome editing in particular and to achieve meaningful impact. Even though Burkina Faso does not currently have a policy or strategy regarding biotechnology, the country has established several frameworks like a biosafety legislation, biosafety regulations, and genome editing guidelines. These instruments are designed to manage the manipulation of biotechnologies, including genetically modified organisms, the use of genome editing, and environmental risks. However, the country urgently needs to develop a national biotechnology and biosafety policy and strategy and must invest

more in agricultural biotechnology and genome editing infrastructure to further improve productivity, innovation and resilience within the sector.

1.3. Engagement in Regional Frameworks

At the regional level, Burkina Faso, alongside the 15 ECOWAS member states and 2 Comité permanent inter-État de lutte contre la sécheresse au Sahel (CILSS) member states (Chad and Mauritania), is required to comply with the ECOWAS Biosafety Regulations, 'C/REG. 04/09/2020,' adopted in September 2020. Decision-making on GEd in West Africa, including Burkina Faso, requires harmonized policies and regulatory frameworks to ensure safe adoption and equitable benefits. The region faces similar challenges—climate variability, low agricultural productivity, pests, and dependence on rain-fed systems—making collaborative research and policy alignment critical. Regional bodies, such as ECOWAS and West African Monetary Union (WAEMU/UEMOA), can play a central role in setting biosafety standards, facilitating knowledge exchange, and supporting joint training programmes that strengthen cross-border capacity. Shared infrastructure, such as regional biotechnology laboratories and gene banks, can also reduce costs while promoting collective food security goals.

1.4. Challenges and Opportunities: Justification of GEd Assessment based on Current Issues Identified in Burkina Faso

1.4.1. Challenges Justifying GEd Assessment

Despite notable progress in laboratory infrastructure and sequencing capacity, several challenges impede the full realization of GEd's potential in Burkina Faso. The significant obstacles hindering the adoption of genome editing in Burkina Faso are the followings:

- **Low investment in research and development:** Burkina Faso's investment in research and development is less than 1% of its gross domestic product (GDP), hindering progress in genome editing and other biotechnology areas.
- **Dependence on external funding and unequal collaborations:** Burkinabè institutions rely heavily on external funding, which can lead to unequal collaborations and limited control over intellectual property and licensing.
- **Gaps in public awareness and stakeholder engagement:** Inaccurate risk perceptions and low levels of public support, often due to misinformation, hinder the adoption of genome editing in Burkina Faso.
- **National policy and strategy:** the lack of national policy and strategy on biotechnology and genome editing, and the lack of school curricula in this field, are a major obstacle to the deployment of biotechnologies and genome editing.

- **Biodiversity conservation:** the use of biotechnological tools, including genome editing, must be reconciled with the protection of biodiversity for the proper safeguarding of ecosystems.

1.4.2. Opportunities Supporting GEd Investment

Biotechnology has the potential to deliver benefits aligned with Africa's needs, rather than being driven primarily by external interests. In Burkina Faso, biotechnology and genome editing offer opportunities to strengthen food security by developing climate-resilient, pest- and disease-resistant, and nutritionally enhanced crops and livestock, as well as improved tolerance to abiotic stresses such as drought, flood, salinity, and low soil fertility (Wambugu & Kamanga, 2014).

The researchers emphasized the potential benefits of genome editing in Burkina Faso. Genome editing offers:

- Greater accuracy and precision: Genome editing enables precise modifications to genomes, reducing the risk of off-target effects;
- Increased efficiency and cost-effectiveness: Genome editing can reduce the time and cost associated with traditional breeding methods;
- Improved crop yields and insect and disease resistance: Genome editing can be used to develop crops with improved yields, pest and disease resistance, and drought tolerance, and thus an adaptation to climate change.

1.5. Objectives of the Landscape Analysis

The main aim of this landscape study was to provide an in-depth assessment of existing policies, infrastructure, institutional capabilities, and technical capabilities to support product development and commercialisation in a select set of African countries.

The specific objectives were to:

- Provide an evidence-based description and analysis of the status of modern biotechnology and GEd in Burkina Faso, highlighting key trends, intervening factors, and areas for intervention, as well as fundamental aspects such as science/technical, political, geopolitical, social, human, cultural, and traditional factors that support or hinder advances in the application of genome editing in agriculture and food systems in Africa.
- Identify the emerging needs that genome editing can readily address, especially those that require rapid responses at scale. These needs will focus on food systems,

including agricultural productivity, reducing postharvest losses, climate adaptation, food and nutrition security, diversified and healthy diets, and

- Identify staple and traditional crops based on African regional contexts that can improve people's livelihoods through food security, better nutrition, climate resilience, and sustainable productivity.

2. APPROACH/METHODOLOGY

Secondary data (literature review) for Burkina Faso were gathered from published literature and institutional websites. In contrast, primary data were collected by a team of trained enumerators who visited key institutions and conducted face-to-face interviews with key stakeholders. In exceptional cases, an online link was used to share the questionnaire. Primary data collected through live interviews were captured using an Open Data Kit (ODK) and stored on the Africa Harvest server. Data from these two sources (primary and secondary) were then analysed, synthesized, and packaged, yielding detailed narratives on the following:

2.1. Status of Policy and Regulatory Frameworks Relevant to Emerging Biotechnologies, Including Genome Editing

Data on components of the regulatory and policy framework in Burkina Faso, collected during the primary and secondary data collection, were retrieved, synthesized, and presented in tables to provide insight into the functionality and preparedness of the country to adopt GEd technologies.

2.2. Projects, Crops, Livestock, Fisheries, Forestry and Traits Ready for Commercialisation And Scaling

As with regulatory and policy frameworks, biotechnology and, particularly, GEd projects, crops, livestock, fisheries, forestry, and traits, as well as the key stakeholders (partnerships) involved and sources of funding, were documented using both secondary and primary data sources. The synthesized and analysed data were used to further:

- i. Identify emerging needs to address economic, social, and environmental/climate benefits.
- ii. Provide information on the status of existing human and infrastructure capacities in GEd technologies in Burkina Faso.

2.3. Staple, Traditional and Commercial Crops Important to Burkina Faso that Could be Improved Using GEd Technology

The data on GEd projects, crops, and traits were further disaggregated into categories based on their highest potential to require GEd technology for national socioeconomic impact and the likelihood of successful completion, considering factors such as national acceptance, resource requirements, and scalability.

2.4. Institutional Capacity (Human Capital, Laboratory and Field Infrastructure, Equipment)

During primary data collection, respondents were asked about existing institutional capacities in human capital, lab and field infrastructure, and equipment for GEd R&D, commercialisation, and scaling (Annex 1). This information was pooled to provide each institution with data on human capital and infrastructure capacity.

2.5. Stakeholder Mapping Relevant for Genome Editing Research-Development and Future Commercialisation (or Deployment)

Targeted sampling was employed, selecting only individuals knowledgeable and currently engaged in modern Agricultural Biotechnology/GEd regulation, policy, R&D, and commercialisation. Some of these key individuals were identified through 1) secondary data, published literature (scientists), 2) databases of institutional websites (regulators and scientists), and 3) referrals through institutional heads (regulators and scientists) or personal knowledge by the country PI. These individuals are spread across the five (5) stakeholder categories identified in the questionnaires (Data collection tools), namely, regulatory agencies, research organizations/institutions, universities, private sector/industry, and government departments/ministries and policymakers. The study employed a mixed-methods approach that combined visits, structured interviews, and questionnaires. Stakeholders were purposively selected to ensure broad representation (Annex 2).

2.6. Database Systems and Database Management

Customized data collection tools (questionnaires) and digital platforms were developed to support primary data collection. Questionnaires were tailored to each stakeholder category and used to generate structured datasets assessing Burkina Faso's preparedness and capacity to adopt, implement, and scale GEd technologies.

Enumerators were identified, recruited, and trained through an online induction programme on the use of ODK and associated data-collection tools. Training was conducted by the IT team and train-the-trainers (ToT) from the consultant. All tools and platforms were pre-tested prior to deployment.

Both methodological and data triangulation approaches were employed to reduce bias and enhance the validity and reliability of findings. All collected data were encrypted and stored on a secure, centralized server to ensure participant confidentiality and compliance with ethical standards.

2.7. Data Synthesis and Data Analysis

Data collected was synthesized and packaged into appropriate information, and where applicable, tables were generated.

Variables investigated included, but were not limited to:

- i. signatory to multinational food and environmental agreements;
- ii. regulatory framework components;
- iii. applications received and approved (field testing, registration, and commercialisation);
- iv. R&D institutions, personnel, and laboratory/field facilities;
- v. projects;
- vi. crops and livestock species; and
- vii. traits under GEd development.

2.8. Interactive Map

An interactive map, comparable to the Agenda 2063 Dashboard, was developed to visualize genome editing activities and capacities across Africa and is accessible at: <https://genome.africa>.

3. RESULTS

3.1. MEMBERSHIP IN MULTILATERAL ENVIRONMENT TREATIES AND AGREEMENTS

Burkina Faso is a Party to the CBD (1993), the Cartagena Protocol (2003), and the Nagoya Protocol (2014), giving it obligations on biosafety, access and benefit-sharing, and the safe movement of living modified organisms (LMOs). It is also a Party to the Nagoya–Kuala Lumpur Supplementary Protocol (liability/redress), to United Nations Framework Convention on Climate Change, to Kyoto Protocol and Paris Agreement (Table 1). Burkina Faso is an active member of the CAC, which is crucial for food risk assessment and the trade of biotech-derived foods. By ratifying these treaties and agreements, Burkina Faso commits to global efforts in areas such as food safety, climate action, biodiversity conservation, and environmental protection, and integrates these obligations into its national food safety and environmental management frameworks as shown in Table 1 below.

Table 1: Status of Burkina Faso's participation in key multilateral environmental agreements

Multilateral Environmental Agreements (MEAs) / Treaties	Date of Ratification / Accession by the Country	Reference
Codex Alimentarius Commission is a joint body of the Food and Agriculture Organization (FAO) and the World Health Organization	Member in 1966	https://www.fao.org/fao-who-codexalimentarius/about-codex/members/en/
Convention on Biological Diversity (CBD)	Ratified 20 Dec 1993	CBD Country Profile – Burkina Faso
Cartagena Protocol on Biosafety	27 Nov 2003	CBD Cartagena Protocol Status
Nagoya Protocol on Access and Benefit Sharing (ABS)	31 Oct 2014	CBD Nagoya Protocol – Parties
Nagoya–Kuala Lumpur Supplementary Protocol on Liability and Redress	21 Mar 2018	CBD Supplementary Protocol
United Nations Framework Convention on Climate Change (UNFCCC)	2 Sep 1993	UNFCCC – Parties List
Kyoto Protocol	31 Mar 2005	UN Treaty Collection – Kyoto Protocol
Paris Agreement	Signed: 22 Apr 2016, Ratified: 11 Nov 2016	UNFCCC – Paris Agreement Status

3.2 Policies that Advance or Hinder Adoption of Modern Biotech and GEd

Burkina Faso three policies/strategies that advance the adoption of modern Biotechnology.

The policies/strategies include:

- **Sectoral policy for agriculture, forestry and livestock (2018-2027):** The policy focuses on boosting productivity, ensuring food and nutrition security, protecting the environment, and building community resilience against climate change and land degradation.
- **National Strategic Investment Plan for Agriculture, Forestry, and Livestock (PNIASP 2021-2025):** The strategic investment plan operates as the overarching framework to transform the nation's agro-sylvo-pastoral sector. In terms of biotechnology, the government has integrated biosafety and genomic innovation into its broader agricultural modernization, climate resilience, and food sovereignty goals (Maahmrah, 2021).
- **National Centre of Scientific and Technological Research (CNRST) Strategic Plan for 2025–2029:** The plan calls for the development of modern biotechnologies in the areas of genetic resource management and the application of genetic engineering (CNRST, 2025).

No policy was found in Burkina Faso during this assignment that hinders or may hinder the adoption of modern biotechnology and GEd.

3.3 National regulatory framework

3.3.1 Biosafety Act/Law, Biosafety Regulations/Decrees and GEd Guidelines

The National Biosafety Agency is the competent authority responsible for general supervision and control over the transfer, handling, and use of genetically modified organisms (GMOs) and for regulating New Breeding Techniques (NBTs), including GEd. Below the components of Burkina Faso's regulatory framework for GMOs and GEd are summarized in Table 2.

ANB began operating according to the Decree No. 262 (2004) before being created in 2005. The biosafety law was subsequently passed in 2006 and renewed in 2012 (Biosafety Law No. 0064-2012/AN (2012)). ANB is the decision-making body and consists of a Governing Board with officials from ten (10) government departments responsible for matters relating to scientific research and innovation, finance, agriculture and animal resources, environment, health, secondary and high education, security, justice, trade, investment, and industry, and four other representatives of governmental and union-related bodies. It is supported in the delivery of its mandate by the central bodies, namely:

The National Scientific Committee on Biosafety (CSNB), which comprises fifteen (15) full members and fifteen (15) alternate members, whose expertise is recognized in the fields of genetic engineering, environmental protection, and human and animal health protection, among others. Members are appointed from 7 ministries and the national ethics committee. The CSNB is an advisory body responsible for: (i) evaluating applications for authorization for the use of genetically modified organisms (GMOs); (ii) proposing corrective measures and validating risk management plans related to the use of GMOs; (iii) assessing risks before, during, and after the development, import, export, testing, or marketing of GMOs and reporting to the National Biosafety Agency (ANB); (iv) assessing economic viability and reviewing compliance with ethical rules.

The National Biosafety Observatory (NBO), which is an advisory body composed of 24 members from nine ministries and civil society. Its missions are to i) to implement a monitoring and surveillance system related to the use of GMOs in health, nutrition, agricultural, environmental, ethical, and their socio-economic impacts; ii) alert the National Biosafety Agency and other relevant authorities of serious risks posed by a genetically modified organism to human or animal health or the environment; iii) assist the National Biosafety Agency (NBA) in its mission to raise awareness and inform/educate the public on biosafety issues within the organizations or associations each member represents; and iv) provide its opinion on the applications submitted to it by the National Biosafety Agency (NBA).

The Scientific and Technical Council (CST), which is made up of 10 members, including three international experts. The mission of the CST is to ensure the quality and consistency of the Agency's projects and programmes, aligned with government objectives for biotechnology and biosafety. If the NSCB is satisfied that the risk assessment and mitigation trials for NBTs or transgenic products are met, it recommends approval of permits to the Director General/CEO of the Agency.

The seed sector in Burkina Faso is regulated by Law No. 010-2006/AN of March 31, 2006, which governs plant seeds. To promote the seed sector, a National Seed Committee (CNS) has been established. The CNS is a consultative and advisory body with 32 members. It is grounded in broad stakeholder participation in the seed sector. The CNS provides advice on scientific and technical issues related to seeds and suggests areas for research. Its membership comprises four from the Ministry of Agriculture, two from the Ministry of Research, two from the Ministry of Forestry, and one from each of the Ministries of Trade, Health, Justice, Territorial Administration, and Animal Resources. The committee also includes members from

professional organizations and specialists in the field. For release purposes, the following are established within the CNS: i) a Subcommittee for the release of Agricultural Varieties (SCHV); ii) a Subcommittee for the release of Basic Forestry Materials (SCHMFB).

Institut de l'Environnement et de Recherches Agricoles (INERA), the lead agricultural research, conducts GEd lab research, such as CRISPR-edited rice for disease resistance, operates under national biosafety frameworks overseen by ANB.

The African Biosafety Network of Expertise (ABNE), launched in February 2010, operates under NEPAD/AUDA support, aids national GEd guideline development and biosafety training.

Long before its first biosafety law, Burkina Faso adopted national biosafety regulations (Règles Nationales en matière de sécurité en biotechnologie) in 2004 to initiate field trials of GM crops, including Bt cotton (BGII), Roundup Ready Flex (RRF) cotton, herbicide-tolerant Bt cotton (RRF-Bt), and Bt maize. In 2006, Burkina Faso enacted a biosafety law, which was revised in 2012 to incorporate the provisions of the Nagoya-Kuala Lumpur Supplementary Protocol on Liability and Redress, adopted in 2010. National Biosafety Guidelines on Gene Editing were developed and validated from 2021 to 2023.

Table 2: Regulatory and institutional landscape for genome editing in Burkina Faso

Institutions	Mandate / Relevance to Ged	Regulatory instruments	Date of enactment or publication	Coverage/ scope	Reference
Agence nationale de biosécurité (ANB)	National biosafety authority; oversees all biosafety and GEd regulatory applications: coordinates risk assessment, public education, and monitoring.	Decree No.–262 (2004); (Biosafety Law No. 0064-2012/AN (2012); Technical Guidance on GEd exemptions (drafted 2023).	Decree: 18 June 2004; Law: 2012; GEd technical guidance: 2023 (under review).	Governs modern biotechnology, including GEd; distinguishes exempt GEd products (no foreign genes, no gene combination)	(PMC , geneconvene.org , NEPAD) https://bch.cbd.int/en/countries/BF/BCP
National Biosafety Scientific Committee (NBSC), Institutional Biosafety Committees (IBCs), National Biosafety Observatory (NBO)	Provide scientific assessment (NBSC), institutional oversight (IBC), and public awareness/monitoring (NBO) for GEd and GM activities.	Embedded in Biosafety Law 2006 and institutional mandates.	Law: March 2006 (Law No. 005-2006/AN)	Support the regulation of GMOs and GE-derived organisms, including confined trials.	(ResearchGate , PMC)
Institut de l'Environnement et de Recherches Agricoles (INERA)	Lead agricultural research; conducts GEd lab research, such as CRISPR-edited rice for disease resistance.	Operates under national biosafety frameworks overseen by ANB.	Ongoing since early experimental approvals (~2005 onward); GEd uses have been reviewed under the 2012 Law.	Conducts confined greenhouse GEd research and pond-based trials (e.g., gene drive labs).	(NEPAD , PMC , geneconvene.org)
African Biosafety Network of Expertise (ABNE)	Regional capacity-building network hosted in Ouagadougou; offers technical support, policy advice, and training.	Operates under NEPAD/AUDA support; aids national GEd guideline development and biosafety training.	Launched: 23 February 2010	Regional centre – supports Burkina Faso and other countries in biosafety/GEd policy and capacity building.	

The approval or licensing process for NBTs or LMOs/GMOs in Burkina begins with an application to the ANB. After the application is checked for completeness and compliance with the provisions of the Biosafety law and its implementing regulations, it is forwarded to NSCB. Members of NSCB are appointed by the Minister of Scientific Research and Innovation and include scientists who are experts in their fields.

This Committee evaluates risk assessments (scientific data on food, feed, and the environment) submitted with each application dossier. The public is also informed and consulted on planned activities related to NBTs or GMOs through notices in major newspapers. Public comments are considered during the evaluation of an application. All potential impacts are assessed, specifically in the context of Burkina Faso, to ensure the sustainability of an NBT or GMO prior to use. Based on the Committee's findings, the application is recommended to the ANB for a decision. This approach promotes credibility and transparency in the regulatory process.

After a product is released, its use is monitored to ensure the assessment conclusions remain accurate. Appropriate public engagement and communication are important to ensure accurate discourse and the sustainable use of these technologies.

The regulatory review processes of biotech products in Burkina Faso are presented in Figure 1, 2 and 3 below.

Decision making process: Notification

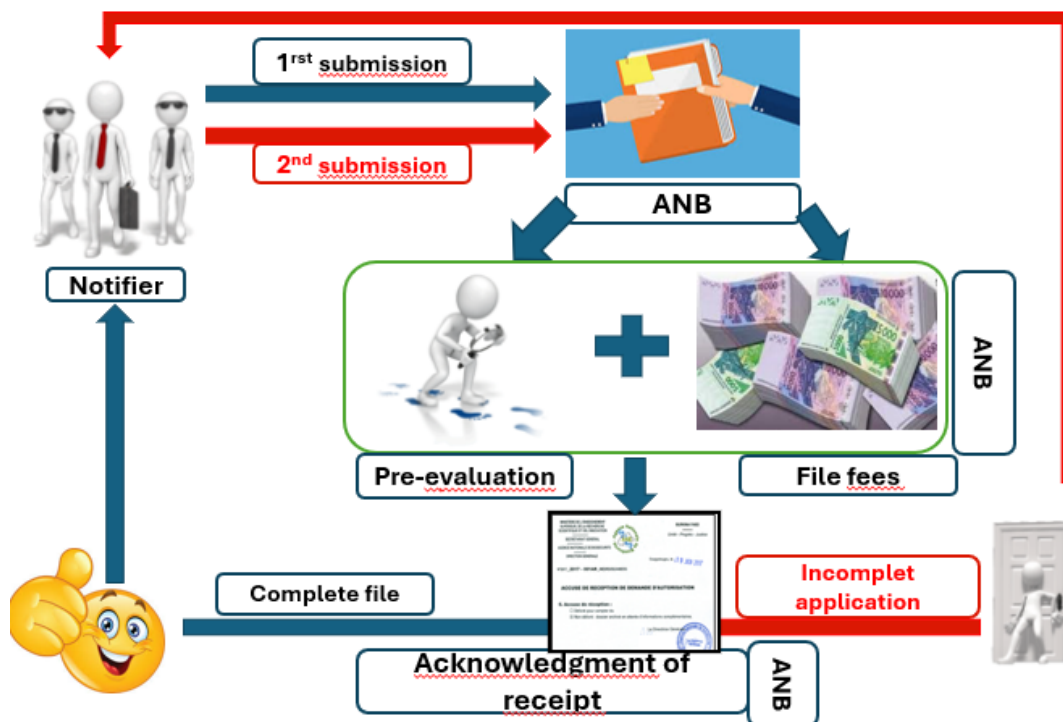


Figure 1: Decision making and notification process for GEd

Decision making process: CFT and Release

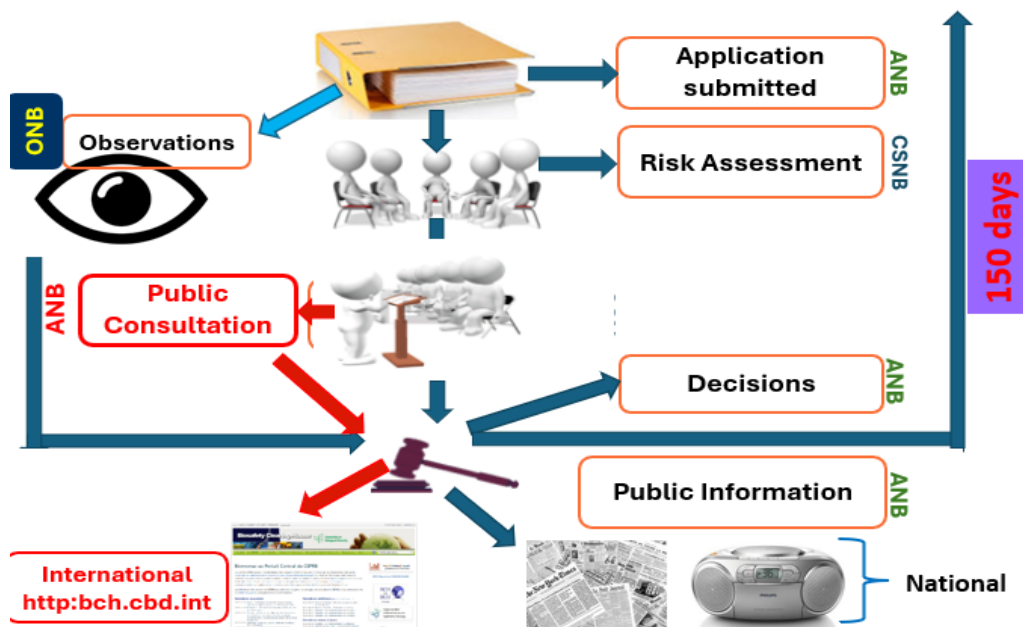


Figure 2: CFT and release

Decision making process: Genome editing

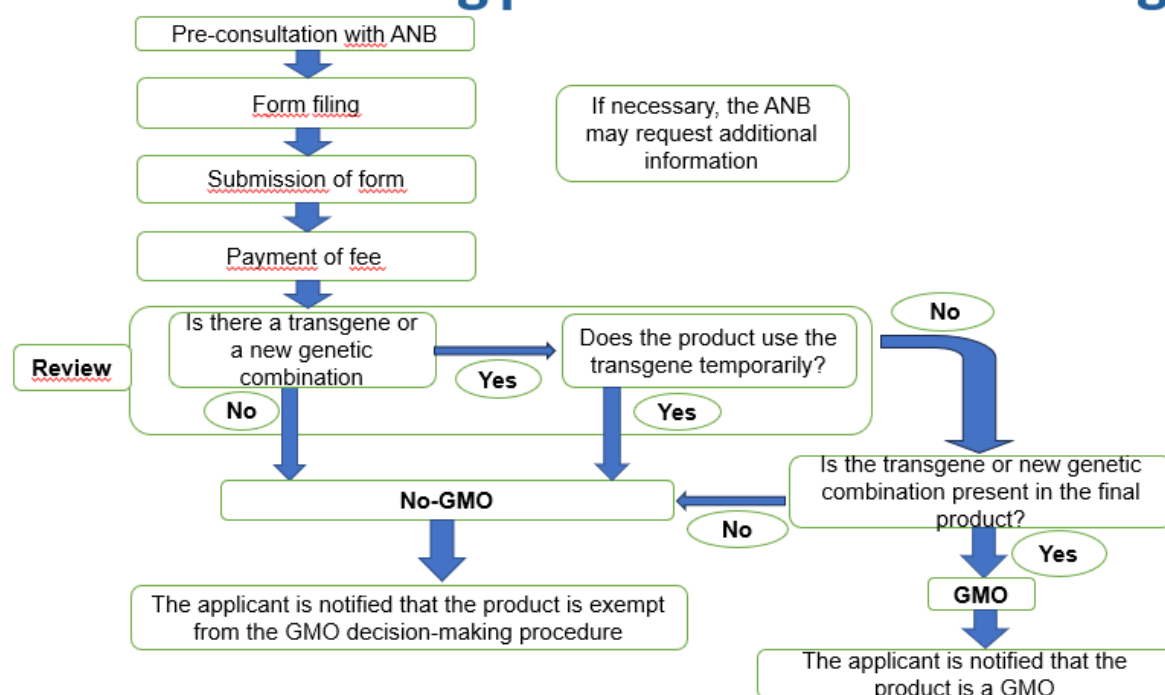


Figure 3: Decision making process for genome editing.

3.4 Socio-economic considerations for decision-making in GEd technology and application

Socioeconomic considerations for decision-making about GEd in Burkina Faso centre on balancing food security, livelihoods, and market access. Rural households depend heavily on agriculture, livestock, and agroforestry for income, making productivity gains from GEd a potential driver of poverty reduction. Key factors include the affordability of improved seeds or breeds, equitable access for smallholder farmers, and safeguards to avoid dependence on costly external inputs. Social acceptance of GEd will depend on awareness campaigns, respect for cultural values, and inclusion of farmers' voices in policy frameworks. Gender equity is also critical, as women play significant roles in seed selection, food processing, and marketing and should directly benefit from new technologies.

From an economic perspective, the main outputs expected from GEd adoption include higher yields of staple crops such as sorghum, millet, maize, and cowpea; improved livestock productivity (milk, meat, eggs); greater resilience of agroforestry systems (shea, baobab, moringa); and higher fishery outputs. These gains could reduce food imports, improve household nutrition, and create opportunities for regional and international trade, particularly in cash crops such as cotton, sesame, and groundnuts. At the same time, cost-benefit

analyses must consider regulatory frameworks, biosafety requirements, and long-term sustainability. A carefully designed GEd strategy could transform Burkina Faso's agri-food systems into a more competitive and climate-resilient sector while maximizing socioeconomic returns for smallholders and the national economy. Socio-economic integration across the region underscores the need to consider trade flows, seed systems, and livestock mobility in the adoption of GEd. Improved crop varieties or livestock breeds developed in one country may benefit other countries, especially when ecosystems and farming systems are similar. However, regulatory misalignment or unequal access to technology could create trade barriers. Thus, regional cooperation in decision-making should prioritize inclusivity, transparency, and respect for indigenous knowledge. A coordinated approach to GEd can enhance West Africa's resilience to climate change, reduce reliance on imports, and strengthen its position in global agricultural value chains.

3.5 Factors that Hinder Application of Modern Biotech and GEd

Political factors: Like in many countries in Africa, the adoption of modern biotechnology in Burkina Faso is dictated by political will. For example, in 2009, the government of Burkina Faso authorized the cultivation of Bt Cotton, whereas in 2016, another government suspended the cultivation of the product. In 2023, ANB renewed BAYER's authorization to grow Bt cotton for another 10 years in Burkina Faso. Governance structures, policy priorities on food security, climate resilience, and poverty reduction, and public trust in science and technology influence Burkina Faso's political landscape regarding GEd. Nevertheless, political will and stability are key for sustained investments in biotech research, regulation, and farmer adoption. Clear decision-making aligned with national agricultural goals will ensure GEd is viewed as a tool for national development rather than an externally imposed technology.

Burkina Faso's historical experience with biotechnology—especially its early adoption and subsequent suspension of GM cotton—offers valuable lessons for GEd governance. Policymakers need to strike a balance between fostering innovation and ensuring biosafety, ethical standards, and public support. Implementing robust regulatory frameworks, ensuring parliamentary oversight, and aligning with regional and continental policies, such as the African Union's biotechnology guidelines, are vital to preventing controversy and building trust. Political backing for research institutions, universities, and farmer groups will play a crucial role in making sure GEd technology promotes inclusive growth, food sovereignty, and national stability.

Geopolitical factors: Most of Burkina's biotech R&D and capacity-building is funded by foreign/external investors (the Gates Foundation, former USAID, AATF, etc.). This creates a situation where modern biotechnology, including GE, is heavily influenced by foreign entities, resulting in a donor-driven approach rather than a purely national one.

Social and Human factors: A significant portion of the burkinabè public fears that modern biotechnology (GMO) products are not safe, particularly concerning toxicity or allergies. Critics of the technology also express widespread concern that its products will harm biodiversity by damaging ecosystems, including beneficial insects and indigenous crop varieties. On seed sovereignty, critics argue that the technology will force smallholder farmers to buy expensive, proprietary seeds from multinational corporations every season rather than save their traditional or landrace varieties.

Misinformation and controversy: In Burkina Faso, a climate of mistrust and the spread of unscientific information are polluting public and political discourse, making it harder for GMOs to gain social acceptance.

Culture and traditions: Certain cultural beliefs and traditions regarding "natural/organic" foods conflict with the concept and application of modern biotechnology in Burkina Faso.

3.6 Genome Editing Programmes and Projects

In Burkina, a GE research project is underway at INERA in collaboration with IRD, AUDA-NEPAD-ABNE, and ANB. This project focuses on developing a solution to bacterial blight in rice, a major challenge for rice production that causes significant yield losses (Table 3). Burkina Faso has over 21 years of experience in modern biotechnology research and has made its mark in the West African sub-region with the commercial cultivation of Bt cotton since 2009. The country continues to make progress in this field, with research on PBR cowpea (open-field trial) and on genetically modified mosquitoes for the sustainable control of malaria. Burkina Faso has developed national guidelines on Genome Editing, which are publicly available on the BCH. These guidelines clarify that Genome editing products that do not contain a transgene are exempt from the process of risk assessment applying to GMOs. Burkina Faso has not yet released any genome-edited crops or products. Table 3 below highlights the literature search on GE projects, organisms (crops), and traits in Burkina Faso.

Table 3: Genome editing projects and programmes in Burkina Faso

Projects/ Programmes	Collaborating partners	GEd Technique	Stage (Lab, field trial, commercialisation)	Funding (US\$)	Funding source	Reference
Rice — Bacterial Blight Resistance	INERA, ANB, AUDA-NEPAD ABNE	CRISPR/Cas9 (gene deletion/edit)	Confined greenhouse trials	86,000	National + AUDA- NEPAD support	(AUDA-NEPAD, 2022 ¹ Akinbo <i>et al.</i> , 2025)
Gene Drive Mosquitoes (Target Malaria project) — Malaria Control	IRSS, Target Malaria consortium (Imperial College, funders)	Gene drive technology	Insectary (lab); planning field release (2025)	Not declared	Bill & Melinda Gates Foundation, Open Philanthropy, EU/UK/NIH, others	geneconvene.org , Suznyk (2024)

¹ <https://www.nepad.org/news/technical-support-development-of-guidance-document-genome-editing-burkina-faso>

Significant obstacles of genome editing in Burkina Faso include 1) low investment in research and development, which is less than 1% of its gross domestic product (GDP); 2) dependence on external funding, and unequal collaborations can lead to unequal control over the products' intellectual property; 3) inaccurate risk perceptions and low levels of public support, often due to misinformation, hinder the adoption of GEd; and 3 lack of a national policy or strategy showing clear policy directions for biotechnology and new breeding techniques, including genome editing, creates significant policy and political uncertainties around decisions on the research, development, and deployment of innovative tools and their products.

Despite these challenges, the researchers emphasized the potential benefits of genome editing, including greater accuracy and precision, which reduce the risk of off-target effects; increased efficiency and cost-effectiveness compared to traditional breeding methods; and the production of crop varieties with improved yields, insect and disease resistance, and drought tolerance.

To harness these GEd potentials, Burkina Faso should promote public awareness and engagement: educating the public about the benefits and risks of genome editing can help build support and trust; fostering international collaborations and partnerships: collaborating with international experts and institutions can help Burkina Faso access knowledge, technologies, and funding; and adopting an enabling policy and strategy that support biotechnology applications for sustainable improvement in agriculture, health, and the environment. According to the respondents, by addressing the challenges and harnessing the opportunities, Burkina Faso can unlock the potential of genome editing and drive innovation in agriculture, health, and other sectors.

3.7 Human Capital and Institutional Capacity

Burkina Faso's agricultural research relies on universities such as Université Joseph Ki-Zerbo, Université Nazi Boni and regional institutions to train scientists, agronomists, and extension workers. These bodies are increasingly offering courses in biotechnology, genetics, and plant breeding, with faculty and students taking international GEd courses. This boosts local expertise, helping researchers improve crops and livestock, but limited lab infrastructure, funding, and equipment hinder large-scale application. Institutions such as INERA and regional centres collaborate with universities and international partners to enhance research capacity for climate and food security solutions (Table 4 and Table 5 below). Better coordination among researchers, policymakers, and extension services is needed to bring innovations to farmers. Investment in training and policy support is vital.

Table 4: Overview of academic and research institutions working on genome editing and related capacity in Burkina Faso

Institution Name	Dept / Unit	GEEd Projects	# of GEEd Researchers	Collaborating Partners	Notable Outputs	Gaps Identified	Reference
INERA (Institut de l'Environnement et de Recherches Agricoles)	Plant breeding/biotech units (various stations)	CRISPR-based rice disease-resistance work (e.g., bacterial blight); broader GM/GEEd crop pipeline under national biosafety oversight	Not publicly disclosed	ANB (biosafety authority), AUDA-NEPAD/ABN E, regional & international crop research partners	Confined greenhouse GEEd research; participation in national GEEd guideline processes	Limited public funding transparency; scale-up capacity (confined trials → multi-location); equipment for high-throughput editing/validation	(PMC , africenter.isaaa.org , USDA Apps)
IRSS / Centre Muraz (Institut de Recherche en Sciences de la Santé)	Medical Entomology & Parasitology (Bobo-Dioulasso)	Gene drive & genetic control of <i>Anopheles</i> (Target Malaria)	Team led by Prof. A. Diabaté; exact headcount not published	Target Malaria consortium (Imperial College et al.), major global funders	Established Target Malaria BF team; insectary readiness; contained-use studies completed	Community engagement & regulatory pathway for open releases; long-term monitoring capacity	(targetmalaria.org , PMC , Gavi)
Université Joseph Ki-Zerbo	BurkinaBioinfo platform; Faculty of Science	Bioinformatics support for genomics/GEEd projects (design, analysis)	Core bioinformatics staff; exact numbers not public	WAVE programme & partners; national wet-labs	National bioinformatics hub established; HPC/bioinfo workflows	Sustained funding, talent retention, and stronger integration with GEEd wet-lab pipelines	(PMC , wave-center.org , ResearchGate)

Institution Name	Dept / Unit	GEEd Projects	# of GEEd Researchers	Collaborating Partners	Notable Outputs	Gaps Identified	Reference
				(INERA/IRSS)	supporting local “wet labs.”		
ANB (Agence nationale de biosécurité)	National Competent Authority for Biosafety	Regulatory guidance for GEEd; early-consultation workflow; risk assessment support	Not applicable (regulator)	AUDA-NEPAD/ABNE; national research bodies	Drafting/validation of GEEd guidance (2021–2023); clarified when specific edits may be treated differently from GMOs	Finalization/operationalization, stakeholder training, and clear product-approval timelines	(NEPAD , nepad-aws.assyst-uc.com , USDA Apps)
AUDA-NEPAD / ABNE (Ouagadougou hub)	African Biosafety Network of Expertise	Technical support & training for GEEd regulation; capacity building	Regional team members are not public	ANB, INERA, IRSS, universities across Africa	National workshops; technical inputs to BF GEEd guidance	Continued funding for training; scaling hands-on regulator training	(nepad-aws.assyst-uc.com)

3.7.1 Women, Youth and Special Interest Groups' Involvement in Modern Biotech and GEd

In Burkina Faso, women, youth, and marginalized farmers are increasingly viewed as essential stakeholders in modern agricultural biotechnology and genome editing. While traditional socio-cultural norms and land tenure systems pose barriers, inclusive governance models are beginning to integrate these groups into decision-making and benefit-sharing.

- **Targeted participatory breeding:** Genome editing and biotech research aim to tailor traits to women's specific needs (e.g., faster cooking times for cowpeas and Bambara groundnuts, higher nutritional value). Participatory selection processes ensure women's preferences are integrated into targeted product profiles.
- **Nutritional focus:** Initiatives, such as Helen Keller International's homestead programmes, have focused on training women to manage small livestock and nutrient-rich crops, serving as a primary entry point for biotechnology and biofortified inputs.
- **Research and development:** Young Burkinabè scientists and researchers are heavily relied upon to translate scientific innovations into actionable solutions. The AUDA-NEPAD and FONRID support systems emphasize capacity building, funding scholarships, and promoting STEM participation for youth and girls to lead modern biotech research.
- **Vulnerable communities:** Groups with disabilities and rural, remote communities are being explicitly targeted in the mainstreaming of Burkina Faso's new climate-resilient frameworks (such as the NDC 3.0), which advocate for equitable access to climate-smart and biotechnology-enhanced crops.

3.7.2 Training and Professional Development

Scientists in Burkina Faso have access to the following GEd courses.

Innovative Genomic Institute: The CRISPR Course is a year-long programme hosted by the Innovative Genomics Institute, in collaboration with the African Orphan Crops Consortium, the Seed Biotechnology Centre at UC Davis, and the International Institute of Tropical Agriculture, as part of the African Plant Breeding Academy. The sessions are conducted in Nairobi, Kenya. Annually, the AfPBA CRISPR Course in Gene Editing trains 10 to 20 doctoral plant scientists across Africa in CRISPR technology. It aims to empower scientists to quickly develop crop varieties tailored to local needs, including climate resilience, improved nutrition, and pest and disease resistance. Selected participants are chosen for their expertise and potential to serve as trainers at their home institutions, where they share CRISPR knowledge with peer trainees.

The following scientists who are based at INERA have been trained in GEd:

1. Dr. Oumar Traore at IRD-France from 2006 to 2008,
2. Dr. Issa Wonni at IRD-France in 2013,
3. Dr. Edgar Valentin Traore at WACCI-Accra-Ghana from 2008-2014 and then at the African Plant Breeding Academy at Nairobi-Kenya in 2016.
4. Dr. Abdalla Dao at the African Plant Breeding Academy in Nairobi, Kenya, in 2024,
5. Dr. Koussao Some from 2008-2014 at WACCI-Accra-Ghana,
6. Dr. Joseph Batieno from 2008-2014 at WACCI-Accra-Ghana

AUDA-NEPAD/OFAB-Burkina: Burkinabe scientists were trained in GEd in a workshop organized by AUDA-NEPAD on biotech solutions in Accra, Ghana, from 25-28 October 2022. Other courses include Biosafety, bioinformatics etc. as outlined in Table 5 below.

Table 5: Overview of Training Programmes on Genome Editing

	Institution / Organizer	Target Audience / # of Trainees per annual	Frequency	Duration	Gaps Identified
Capacity building in genome editing & biosafety regulation	ANB (National Biosafety Agency) with AUDA-NEPAD/ABNE	Regulators, researchers, policymakers (~20–30 annually)	Periodic workshop (every 1–2 years)	3–5 days	Limited continuous training; weak integration with product developers; lack of hands-on lab focus
CRISPR-Cas genome editing workshops (rice & staple crops)	INERA + regional partners (e.g., AfricaRice, WAVE programme)	Plant breeders, molecular biologists (~15–20 per cycle)	Ad hoc, project-based	1–2 weeks	Limited lab infrastructure; reliance on external trainers; no sustained curriculum
Gene drive & vector control training	IRSS (Institut de Recherche en Sciences de la Santé) – Target Malaria	Entomologists, molecular biologists, technicians (~10–15 annually)	Annual, depending on project phase	Several weeks to months (insectary and molecular work)	Specialized focus (mosquitoes only); no spillover to agricultural GEd
Bioinformatics & genome analysis for editing projects	Université Joseph Ki-Zerbo – BurkinaBioinfo platform	Students, junior researchers, data analysts (~20+ per intake)	Annual short courses, modules in degree programmes	1–2 weeks (modules)	Need stronger linkage to wet-lab genome editing and validation capacity
Regional hands-on GEd training (Burkina Faso participants)	RUFORUM, WAVE Programme, Africa Biosafety Network of Expertise	Early-career researchers (numbers vary; BF trainees included)	Annual or bi-annual regional trainings	1–2 weeks	Limited local hosting in Burkina Faso; trainees often must travel abroad

In addition, several Burkina Faso higher institutions of learning, particularly universities, are offering training courses (Certificates, Diplomas, BSc, MSc, and PhD degrees) in biotechnology and its related disciplines, which can be major drivers and leverage for GEd technologies (See Table 6).

Table 6: Universities Offering training in Biotechnology and GEd-related courses in Burkina Faso

Institution	BSc Program	Master's Program	PhD Program
University Joseph Ki-Zerbo	General biology Quality Management in Agro-food Industries	Molecular Biology & Genetic Engineering & Biotechnology Quality Management in Agro-food Industries Biodiversity and characterization in animal production	Molecular Biology & Genetic Engineering & Biotechnology Quality Management in Agro-food Industries Biodiversity and characterization in animal production
University Nazi BONI of Bobo-Dioulasso	Biological Engineering	Molecular Biology & Genetic Engineering Parasitology–Mycology– Medical & Veterinary Entomology	Molecular Biology & Genetic Engineering Medical/Veterinary Entomology & Ecology

3.8 Infrastructure and Equipment

3.8.1 Universities and Research Institutions

The studies assessed the status infrastructure and equipment including the containment-level facilities across the selected institutions. The results of this assessment are summarized in Table 7 below. Core infrastructure required for genome editing research and development including laboratories, greenhouses, and field trial farms is available across all selected institutions.

Containment levels for genetically modified organisms (GMOs) are defined by the degree of physical and biological safety measures required to manage potential risks associated with handling such organisms. These containment levels range from Level 1, representing minimal

or negligible risk, to Level 4, which applies to high-risk organisms requiring maximum isolation. The criteria for each containment level are outlined below:

- Containment Level 1 (CL1): Applicable to GMOs with no or negligible risk. Facilities require basic laboratory infrastructure, standard operating procedures, and general hygiene practices.
- Containment Level 2 (CL2): Applicable to low-risk GMOs. Facilities require primary containment measures such as biological safety cabinets or fume hoods, controlled access, and enhanced laboratory practices.
- Containment Level 3 (CL3): Applicable to moderate-risk GMOs. These facilities involve specialised laboratory design, directional airflow, restricted access, and mandatory use of personal protective equipment.
- Containment Level 4 (CL4): Applicable to high-risk GMOs. Facilities require maximum biological isolation, dedicated buildings, airlocks, advanced decontamination systems, and highly specialised operational protocols.

Most Burkina Faso universities and research institutions possess basic molecular biology laboratories and greenhouse facilities. However, these institutions face constraints in upgrading their sequencing services, funding, shortage of genome-editing experts and procurement processes, which limit their ability to fully support genome-editing research (See Table 6).

Table 7: Status and needs assessment of biosafety laboratory facilities by institution.

Institution	Type of Facility	Biosafety Level	Status	Limitations	Support Needed
INERA (Institut de l'Environnement et de Recherches Agricoles)	Plant biotechnology labs (Ouagadougou, Farako-Bâ)	Containment Level (CL) CL-1/CL-2	Operational, used for molecular biology and plant transformation	Limited capacity for CRISPR work; insufficient high-containment greenhouses	Upgrade of equipment (CRISPR platforms, growth chambers); training in molecular diagnostics
IRSS / Centre Muraz (Bobo-Dioulasso)	Insectary & molecular entomology labs (Target Malaria)	CL-2 (insect containment facilities)	Operational, supports gene drive R&D under strict containment	Restricted to entomology; limited expansion for broader genome editing work	Strengthened monitoring, molecular verification tools, and long-term funding for biosafety compliance
Université Joseph Ki-Zerbo	Molecular biology & bioinformatics labs (BurkinaBioinfo)	CL-1 / CL-2	Functioning: supports genomics and computational biology	Limited wet-lab genome editing capacity; infrastructure gaps for plant/mammalian cell editing	Investment in genome editing platforms; integration with bioinformatics
National Biosafety Agency (ANB)	Regulatory testing and biosafety monitoring labs	CL-1	Basic facilities for sample handling & GMO detection	No advanced genome editing detection platforms; reliant on external reference labs	Strengthening analytical labs (NGS, CRISPR detection); training of biosafety inspectors
Regional / Partner-supported labs (e.g., WAVE Programme)	Plant pathology/virus detection labs	CL-2	Operational, supports West African crop virology research	Mostly focused on virus diagnostics, not full GEd workflows	Expansion to genome editing and functional validation platforms

3.9 Traditional and Staple Crops, Livestock, Agroforestry, and Fisheries Varieties/ Breeds for Improvement Using GEd

Burkina Faso's agriculture is dominated by traditional and staple crops such as sorghum, millet, maize, rice, and cowpea, which are central to food security but face low yields, recurrent drought, and pests. Traditional varieties often show strong adaptation to local soils and climates but lack the productivity and resistance traits needed for sustainable growth under climate change. Groundnuts, sesame, and cotton are also important cash crops, while shea and baobab trees in agroforestry systems provide nutrition and income. Genetic improvement through modern breeding and gene editing could enhance drought tolerance, pest resistance, and nutritional value while conserving the resilience of local landraces. Table 8 below highlights commercial and traditional crops amenable to GEd improvement based on Burkina Faso Policy Framework for Investment in Agriculture (OECD, 2013).

Livestock production, including cattle, goats, sheep, and poultry, plays a vital role in livelihoods and nutrition but is constrained by diseases, feed shortages, and limited genetic diversity. Traditional breeds are well adapted to harsh conditions but often have low productivity in milk, meat, and egg yields. Agroforestry systems integrate livestock, crops, and trees, strengthening resilience. Fisheries primarily artisanal and seasonal provide protein but remain underdeveloped. Using GEd to improve local breeds, diversify agroforestry species, and enhance fish strains could increase productivity and food security. Strategic investments in genetic resources, while respecting indigenous knowledge and biodiversity, would be key to the country's sustainable agricultural transformation.

Table 8: Priority organisms for genome editing application

Crops/Livestock /Agroforestry/Fisheries	Trait of interest improved	Socio-Economic Justification	GE Potential (Low/Medium/High)	Existing R&D	Actual vs Expected Annual Production Capacity (tonnes)
Sorghum (crop)	Drought tolerance; Striga & pest resistance; improved nutritional quality	Staple for food security in Sahelian zones; widely grown by smallholders resilient sorghum stabilizes diets & incomes.	High single-gene edits and regulatory/physiological targets feasible; speed > conventional breeding for some traits.	National breeding programmes (INERA) + regional CGIAR work on sorghum genetics.	Actual: ~1.8–1.9 million t/year (recent series). Expected: incremental yield gains targeted by national programmes. (FAOSTAT, 2024)
Millet (crop)	Drought tolerance; pest resistance; improved nutritional profile	An important staple in marginal areas; supports food security and climate resilience.	High, similar to sorghum, feasible for key tolerance traits.	National research + regional millet initiatives; limited published GED projects locally.	Actual: ~0.9 million t/year (recent). Policy aims to maintain/raise productivity. (FAOSTAT, 2024)
Maize (crop)	Drought tolerance; insect/pest resistance; quality protein maize,	Growing staple and feed crops yield stability reduces food insecurity and market volatility.	High maize is a model crop for GED; many global precedents.	INERA & regional CGIAR (CIMMYT) collaborations; national seed systems.	Actual: ~1.85 million t/year (recent 5-yr averages). Targets aim for modest increases using improved varieties. (FAOSTAT, 2024)
Rice (crop)	Disease resistance (bacterial blight), yield, stress tolerance.	Strategic crop for urban diets and import substitution — improved rice reduces import bill and raises incomes.	Medium–High — CRISPR targets demonstrated globally; local transformation/regeneration capacity required.	INERA GED rice work (CRISPR, greenhouse trials) noted in national programmes.	Actual: ~0.5 million t/year (recent). National policy aims to increase domestic rice production. (FAOSTAT, 2024)
Cotton (cash crop)	Insect resistance (bollworm), herbicide tolerance, fibre quality	Major export & rural cash crop (livelihoods, foreign exchange). Past adoption of Bt cotton shows private-sector linkages.	Medium — transgenic approaches historically used; GED could refine traits (less regulatory hurdles if SDN-1/2 exempted).	Historic BT cotton programmes; INERA & private sector seed companies.	Actual: cotton seed unginned, 770 000 T; production is important for export revenue. (FAOSTAT, 2024)

Crops/Livestock /Agroforestry/Fisheries	Trait of interest improved	Socio-Economic Justification	GE Potential (Low/Medium/High)	Existing R&D	Actual vs Expected Annual Production Capacity (tonnes)
Cattle/dairy (livestock)	Disease resistance, heat tolerance, improved milk yield/ composition	Livestock underpin nutrition, livelihoods, and value chains; dairy production expanded to increase household protein and income.	Medium — some monogenic targets are feasible (disease susceptibility); polygenic yield traits are more complex.	Veterinary & livestock programmes (Ministry, research services); limited local GEd documented.	Milk (historical): ~190 million litres (2013 data) with growth trends since; formal slaughter/processing increasing. (FAO, 2026, The Farmer's Journal, 2026)
Poultry (livestock)	Disease resistance (Newcastle, IAV), growth/feed conversion, heat tolerance	Rapid income generation for smallholders; high urban demand for poultry meat & eggs.	High — targeted edits for disease resistance/immune factors are feasible and have a high impact.	National poultry sector actors; limited published local GEd projects — strong potential for private sector uptake.	Actual: poultry meat production in the tens of thousands t/yr (projected ~49k t in 2021; ~53k t by 2026). (ReportLinker)
Pigs (livestock)	Disease resistance (ASF), fertility, and feed efficiency	Growing demand for pork; ASF is a significant constraint — resistant stock would transform smallholder value chains.	Medium — ASF resistance challenging but high-impact if achieved; fertility edits are more tractable.	National pig programmes; no public GEd pig projects reported.	Actual: sector statistics limited; pig meat contributes a modest share vs poultry/beef.
Tilapia / inland fisheries & aquaculture (fisheries)	Growth rate, disease resistance, sex-ratio control	Freshwater fisheries are important for nutrition; aquaculture can reduce imports and create jobs.	High — tilapia is a common GEd target globally (growth, disease resistance, sex control).	Small but growing aquaculture sector; regional R&D and FAO/World Bank support for pond systems.	Fisheries (total): ~31,000 t (2022, total fisheries production); aquaculture is small (hundreds of tonnes) but has growth potential. (ResearchGate , Open Knowledge FAO)
Agroforestry/fuel wood & timber species (e.g., Eucalyptus, indigenous trees)	Faster growth, pest/disease resistance, wood quality, root architecture (erosion control)	Agroforestry underpins fuel, pole, soil conservation, and carbon projects—important for hillside management and rural energy.	Low–Medium — GEd possible but long generation times slow impact; useful for pest resistance or wood property edits.	Forestry extension, nurseries, national forestry programmes; little/no GEd tree work locally.	Production metric: typically area/volume (m³) rather than tonnes; national afforestation/planting campaigns are ongoing.

3.10 Intellectual Property Rights and Benefit Sharing

Burkina Faso is a member of the African Intellectual Property Organization (OAPI) and the World Intellectual Property Organization (WIPO), which provide access to registration systems for trademarks, patents, and industrial designs (Table 9). The National Centre for Industrial Property (CNPI) is the official body in Burkina Faso responsible for promoting and protecting innovations, trademarks, and patents. It serves as the national liaison for the African Intellectual Property Organization (OAPI).

Table 9: Burkina Faso's IPRs, legal instruments (laws) and organizations in charge

Intellectual Property Rights	Organization / Institution	Laws / Regulations / Protocols / Treaties
Trademarks, Patents, and Industrial Designs.	African Intellectual Property Organization (OAPI)	Bangui Agreement of March 2, 1977
	World Intellectual Property Organization (WIPO)	The Paris and Berne Conventions (1967)

3.11 Private Sector Participation

The involvement of Private Sector seed companies in the commercialisation of genetically modified (GM) crops in Burkina Faso is a crucial aspect of the country's agricultural development. The private seed companies have been instrumental in the commercialisation of GM crops, particularly Bt cotton (Table 10).

The international seed companies with representatives in Burkina Faso include.

- **Bayer:** Developed the GM cotton event Mon 15985 released in 2008, renewed in 2023 for the next 10 years, and donated the Bt genes for PBR Cowpea.
- **Seedwork International Limited** is a seed research and development company engaged in research, production, and marketing of hybrid seeds. SEEDWORK is engaged in demonstration trials on Bt-Hybrid cotton in Burkina Faso.
- **MAHYCO (Maharashtra Hybrid Seeds Co.)** is an agricultural company based in India and a major seed producer. Mahyco is conducting demonstration trials of Bt-Hybrid cotton in Burkina Faso.
- **Nankosem** focuses on the production, importation, and distribution of vegetable hybrid seeds and other agricultural products. Its customer base consists entirely of smallholder farmers.
- **Prophyma SA** offers a wide variety of highly resistant hybrid fruit and vegetable seeds.
- **SEMAFORT was** established in 2017 in Bobo Dioulasso with a mission is to conduct research, produce, and market high-quality seeds and their parent lines across ECOWAS countries.

The local seed companies are small and struggle to survive. Furthermore, the diversification of seed suppliers poses the challenge of coordinating and leveraging their specific strengths. However, there are signs of structuring into networks or associations to better interact with other actors and stakeholders.

Private seed companies have organized into a union and an active seed trade association to advocate for their interests. The National Union of Seed Producers' Cooperative Societies of Burkina Faso (UNPSB), founded on November 20, 2004, with the support of the government and its founding members, is a recognized seed producer network. Officially recognized by decree No. 2004-000190 / MATD / PKD / H on December 23, 2004, it functions as a professional organization with economic and social roles, based on cooperative principles. Its members include Regional Unions of Seed Producers' Cooperatives (URPS). UNPSB links approximately 3,500 seed producers through branches organized as SCOOPS at the local level and as cooperative societies at the provincial/regional level, and is governed by a Board.

The organization promotes food security by producing improved seeds and aims to increase yields, supporting members' livelihoods. The National Association of Seed Companies of Burkina Faso (ANES-BF) was established in 2017 and comprises 13 companies, some of which are well structured. Its mission is to promote the full adoption of agricultural technologies. INERA, using its basic seeds, aims to transform Burkina Faso's agriculture into a highly productive, efficient, competitive, and sustainable system to ensure food security and lift the country's 13 regions out of poverty. The government purchases seeds from seed producers organized through the UNPSB and ANES-BF and redistributes them at subsidized prices to vulnerable populations. The union and the association are also seed producers:

Neema Agricole du Faso SA (NAFASO), founded in 2008, sells improved maize, sorghum, and cowpea seeds and produces at least 5,000 tons per season. It has distribution network of 50 stores and 450 resellers in more than 11 countries in West and Central Africa.

Faso Agriculture and Inputs (FAGRI) provides high-quality improved seeds to its customers, offers services and builds trust. It deals in sunflower, soybean, cassava, potato, and vegetable seeds (okra, onion, tomato, etc.), forage seeds (Mucuna, Dolichos, pigeon pea), forest seeds (Prosopis, Acacia, etc.), and herbaceous seeds (Panicum, Andropogon, etc.).

EPAM (Enterprise and Promoter of Modern Agricultural Products), founded in 2012, specializes in the production and marketing of vegetable and cereal improved seeds.

LAVODEC (La Voix des Champs) specializes in the production and marketing of improved agricultural seeds.

Table 10: Overview of private sector companies involved in seed production & distribution, crop improvement & biotech/GEEd activities in Burkina Faso

Company/Entity	Type (Agri-biotech, Start-up, etc.)	GEEd Activities	Partnerships	Challenges Faced	Investment Interest
Bayer CropScience	Multinational	No GEEd	Pannar, Faso Kaba, AGRA	Regulatory and seed certification hurdles, dominance of informal farm-saved seeds; seed counterfeits; high cost of seed production, climate change	Biotech R&D (Bt cotton), Bt cowpea gene donation; conventional seed breeding
SEEDWORKS		No GEEd	Novalliance (subsidiary Nankosem), Syngenta Foundation (Seeds2B), CIMMYT		Biotech R&D (Bt cotton field demo); conventional seed breeding
MAHYCO			Quton Seed Co.		
SEMAFORT	Regional	No Biotech/GEEd	NAFASO, LAVODEC, PROPHYMA SA		Conventional seed breeding, seed production and distribution
PROPHYMA SA	National				
Nankosem	National	No GEEd	Technisem, Tropicasem, Seed Co West and Central Africa, Agro-dealers		Importation, multiplication and distribution of vegetable seeds

3.12 Funding and Investment Landscape

The primary funding source for modern biotechnology and GEEd research in Burkina Faso is the Gates Foundation. Other sources include competitive international grants from entities like the European Union and various trusts, alongside funding from the Industrial Development Corporation. Table 11 below outlines potential agencies that could support GEEd technology implementation.

Table 6: Overview of national and other funding sources for genome editing

Funder/Donor	Organization Type	GEd Project	Amount (USD)	Duration	Recipient Institution(s)	Area of Focus
The Gates Foundation	Philanthropic foundation	Core funder of Target Malaria (genetic control of <i>Anopheles</i> mosquitoes); also major donor to AUDA-NEPAD/ABNE biosafety activities supporting GEd guidance	Target Malaria (consortium total): BMGF ~\$155.5M (part of ~\$173M total fundraising 2005–2025); BMGF grants to ABNE/NEPAD (phase grants); part of multi-million packages	Multi-year (Target Malaria: 2005–2025; ABNE grants across 2010s)	Imperial College (lead of Target Malaria), IRSS (BF partner), AUDA-NEPAD/ABNE, national agencies (ANB, INERA)	Gene-drive & GMO mosquito research; biosafety/regulatory capacity building. (targetmalaria.org , PMC)
Open Philanthropy	Philanthropic fund	Core funder (co-funder) of Target Malaria research & preparedness	~US\$17.5M (part of Target Malaria funding portfolio 2005–2025) — country disbursements not specified	Multi-year (supporting consortium activities across 2000s–2020s)	Imperial College / Target Malaria consortium; in-country partners (IRSS)	Gene-drive research; stakeholder engagement and regulatory readiness. (targetmalaria.org)
AUDA-NEPAD / African Biosafety Network of Expertise (ABNE)	Regional development agency/capacity network	Technical support & capacity building for national GEd guidance and biosafety (workshops, guidance docs for BF)	Phase II funding (incl. Gates supplement): total Phase II ≈ US\$11.93M (supplement ≈ US\$1.489M); Burkina-specific share not broken out	Ongoing/regional (2010s → present); specific GEd guidance support in 2021–2023	ANB (Burkina), INERA, IRSS, national regulators	Biosafety capacity building; development of national GEd guidance documents. (NEPAD)
European Union / UK / Other public donors	Multilateral/bilateral donors	Co-funding and research support for Target Malaria and related public-health GEd preparedness (grants to consortium	Not publicly disaggregated by country (consortium-level support aggregated)	Varied multi-year grants to research consortia	Imperial College (lead), partner institutions, including IRSS	Vector control research, regulatory & community engagement preparedness. (targetmalaria.org , Gavi)

Funder/Donor	Organization Type	GEEd Project	Amount (USD)	Duration	Recipient Institution(s)	Area of Focus
		partners and regional activities)				
National government (Burkina Faso)	State / public funds	Core recurrent funding and in-kind support for ANB, INERA, and IRSS (enabling confined GEEd research, insectary operations, and regulatory activities)	Not publicly disclosed (budget lines in national science/health/agriculture allocations)	Annual national budget cycles; project durations vary	ANB (Agence Nationale de Biosécurité), INERA, IRSS, Université Joseph Ki-Zerbo	In-country research operations, regulatory review, and implementation support for trials. (PMC , USDA Apps)
Project/programme grants (regional, e.g., GIZ, EU NEPAD grants)	Bilateral/regional development agencies	Infrastructure & capacity grants (e.g., AU infrastructure support, NEPAD/GIZ grants for regional projects that benefit BF biosafety/GEEd readiness)	Examples: NEPAD/GIZ and other grants in € millions (country-level shares not continuously published)	Project-based (multi-year)	ANB, AUDA-NEPAD hubs, national research institutes	Infrastructure, training, and regulatory systems strengthening that support GEEd readiness. (NEPAD)
Research donors & CGIAR / foundations (indirect support)	Research & philanthropic bodies	Support to crop research and capacity building (e.g., INERA collaborations on CRISPR rice lines supported through CGIAR/regional programmes)	Not publicly disaggregated (project grants vary)	Project-based	INERA, regional CGIAR centres, and universities	Crop GEEd research (rice, cotton trials), training, and germplasm exchange. (USDA Apps , nepad-aws.assyst-uc.com)

Institutions visited and categories of stakeholders engaged:

Research institutions and academia: These included UJKZ in Ouagadougou, UNB in Bobo-Dioulasso, INERA, and the Institute of Research in Health Sciences. The other institutes visited were the African Centre of Excellence in Biotechnological Innovations for the Elimination of Vector-Borne diseases (CEA/ITECH-MTV), the National Institute of Public Health (INSP), the Pietro Annigoni Biomolecular Research Centre (CERBA), the Health Action Research Group (GRAS).

Government & regulatory authorities: The following were visited: ANB, the National Seed Committee (CNS), the Subcommittee for the release of Basic Forestry Materials (SCHMFB), and the Subcommittee for the release of Agricultural Varieties (SCHV).

Other organizations: Also visited were private companies, including Bayer, UNPCB, ANES-BF, LAVODEC, EPAM, FAGRI, NAFASO, and SEMAFORT. The not-for-profit bodies visited were the OFAB Burkina Chapter, the African Society of Genetics, RD-STIA, and media networks. Funding Organizations visited included the Africa Agricultural Technology Foundation (AATF), FONRID. The list is presented in Annexure 2 below.

4. RECOMMENDATIONS AND POLICY OPTIONS

4.1. Recommendations

Strengthen domestic funding: To reduce reliance on external donors, Burkina Faso should significantly increase government and private-sector investment in genome editing (GE) research. A National GE Fund, anchored in the Ministry of Higher Education and Scientific Research, could ensure long-term, predictable financing. Blended models that combine public allocations, competitive grants, and tax incentives for private R&D would create a sustainable ecosystem. Priority areas should include staple crops, livestock health, agroforestry species, and public health challenges.

Build human capacity: Developing human capital is central to GE's success. Burkina Faso should establish state-of-the-art laboratories, sequencing centres, and bioinformatics hubs in Ouagadougou, Bobo-Dioulasso, and other research zones. Training programmes should include MSc and PhD fellowships, technician apprenticeships, and return-scientist packages to reduce brain drain. Field phenotyping networks and extension-scientist exchanges will ensure that research remains grounded in farmers' realities.

Enhance regulatory oversight: The National Biosafety Agency must expand its scope and technical expertise to effectively govern emerging genome-editing technologies. A tiered, product-based regulatory framework should streamline approvals for edits that do not introduce foreign DNA while maintaining strict oversight of complex modifications. Ethics and indigenous knowledge boards should guide decision-making, and digital one-stop portals can improve transparency and accountability.

Public engagement and communication: Public trust in GE depends on inclusive communication strategies. Outreach should include radio dramas, community meetings, farmer field schools, and materials in local languages. Training journalists, extension agents, and community leaders to serve as trusted messengers will help counter misinformation. Participatory platforms, such as citizen juries and co-design workshops, should give farmers, women, and youth direct input into technology priorities.

Promote regional and global partnerships: Given the shared agricultural and health challenges in West Africa, Burkina Faso should foster cross-border collaborations in GE research. Alignment with ECOWAS and WAEMU biosafety regulations would facilitate technology transfer and regional trade. Partnerships with CGIAR centres, African Biosciences networks, and the African CDC can expand research opportunities. Joint training and mobility schemes will strengthen regional talent pipelines while promoting fair benefit-sharing.

Integrate genome editing into national policy: Genome editing must be mainstreamed across Burkina Faso's agricultural transformation, health innovation, and climate resilience

strategies. Curricula from secondary schools to universities should include GEd, bioinformatics, and ethics. National data policies should ensure that genomic data are stored, shared, and governed responsibly. To ensure sustainability, equipment procurement should be accompanied by local maintenance contracts and support for spare parts.

Cross-cutting equity and access measures: A comprehensive GEd strategy should include gender and youth equity targets, ensuring that at least 40% of trainees and project leaders are women or young scientists. Farmers' access to improved seeds and breeds must be safeguarded through royalty-free or low-cost licensing tiers. Environmental monitoring should include refugia, biodiversity protection, and post-release assessments to minimize ecological risks while maximizing benefits.

Implementation roadmap: In the short term (0–12 months), legal updates for ANB, establishment of budget lines, and launch of public awareness campaigns should be prioritized. Within one to three years, Burkina Faso should operationalize regional labs, initiate field trials under tiered regulation, and adopt harmonized regional standards. Within three to five years, certified GEd crop varieties, livestock breeds, and health applications should be released and scaled through farmer networks and seed/breed systems.

Governance and monitoring: A National Genome Editing Council comprising government, research institutions, farmers, private-sector actors, and civil society should oversee the implementation. The Council should publish an annual State of Genome Editing Report with clear performance metrics, compliance audits, and recommendations. Independent evaluations every two to five years will ensure accountability and guide adjustments. By embedding transparency, inclusivity, and strong governance, Burkina Faso can build a GEd framework that advances food security, health, and economic resilience.

4.2. Policy Options

- **Short-term:** (12-24 months): communication and public engagement-managing the on-going misinformation, disinformation and mal-information surrounding GMOs is also affecting GEd.
- **Medium-term (2–5 years):** Include leveraging biotech for national development, and trade in biotech products while ensuring compliance with international standards. Fostering sustainable agricultural practices, improved food security, and promoting economic growth through advances in Biotech, especially GEd techniques and regional standing
- **Long-term:** (5-10 years): Regional and international collaborations, sustained investment in genome editing R&D.

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ANNEXES

Annexure 1: Criteria for Determining Laboratory Status for BSL-1 and BSL-2 Operations

- (i.) **Criteria for infrastructure and equipment for CL1:** 3-4 rooms containing the following: PCR, Incubator, Sequencers, Freezers (-80, -20), P/ATC room, Access to consumables, LAF chamber, Electrophoresis Apparatus, Autoclave, Microwave, Vortexer, UV illuminator
- (ii.) **Criteria for infrastructure and equipment for CL 2:** Standard Microbial Practices + Special practices + All CL-1 equipment plus a mandatory biosafety hazard sign, special protective gear, special Cabinets (class II), controlled access to rooms, etc., handling agents of moderate potential hazards to people + animals + environment

	Conditions	Status
CL 1	If all in (i) above are available with or without the sequencer	Fully equipped
	Missing any of the other equipment in addition to the sequencer	Not fully equipped
CL 2	Conformance to the criteria in (ii) above	Fully equipped
	Any non-conformance to the criteria in (ii) above	Not fully equipped

Annexure 2: Institutions Visited and Key Resource Persons Consulted

The table below provides an overview of institutions visited, stakeholder categories, and key resource persons who provided data and information during the Genome Editing Landscaping Survey in Burkina Faso.

Category	Institution / Organization	Location	Key Resource Persons	Designation
Research & Academia	National Centre of Scientific and Technological Research (CNRST)	Ouagadougou	Dr. Emmanuel Nanema	General Delegate
Research & Academia	Institute of Environment and Agricultural Research (INERA)	Ouagadougou	Dr. Drissa Sérémé	Director
			Dr. Amadou Traore Dr. Arnaud Tapsoba Dr. Fabiola Traore	Head of Unity Scientists (Laboratory of biology and animal health, Unity of animal genetics)
			Dr. Issa Wonni	Scientist (Rice and rice growing programme/ Edited rice Project)
			Dr. Ezekiel Tibiri Dr. Monique Soro	Scientists (Laboratory of Virology and Plant Biotechnology)
			Ir. Leandre Poda Ir. Serge Felicien Zida	Scientists (Laboratory of Genetic and Plant Biotechnology)
		Bobo-Dioulasso Bobo-Dioulasso	Dr. Eric Sombie	Coordinator of the National Centre of Specialization on fruits and vegetables
			Dr. Dieni Zakaria	Scientist (Cotton and Textile Fiber Programme/ Laboratory of Genetics)
Research & Academia	Institute of Research in Health Sciences (IRSS)	Ouagadougou	Dr. Lazare Belemnaba	Director
		Bobo-Dioulasso	Dr. Moussa Namountougou	Scientist (African centre of excellence in biotechnological innovations for the elimination of vector-borne diseases (CEA/ITECH-MTV)/Target Malaria)
		Ouagadougou	Prof. Jean François Kobiane	President

Category	Institution / Organization	Location	Key Resource Persons	Designation
Research & Academia	University Joseph KI-ZERBO (UJKZ), Ouagadougou		Dr. Florencia Djigma	Scientist (Laboratory of Molecular Biology and Genetic Engineering (LABIOGENE), UJKZ)
			Prof. Nerbewende Sawadogo Dr. Romaric Nanema	Scientists (Biosciences Laboratory, UJKZ)
			Prof. Mamoudou Dicko	Head of Laboratory of biochemistry, biotechnology, food and nutritional technology (LABIOTAN)
Research & Academia	National Public Health Institute (INSP)	Ouagadougou	Dr. Barro Seydou Dr. Wilfried Ouedraogo Prof. Isidore Bonkounbou	Director General Scientific Director Technical Director (Central Laboratory of Reference/INSP)
Research & Academia	University Nazi BONI (UNB), Bobo-Dioulasso	Bobo-Dioulasso	Prof. Hassan Bismarck Nacro	President
Government & Regulators	National Agency of Biosafety (ANB)	Ouagadougou	Dr. Koussao Some Ir. Mathurin Rouamba Ir. Mounyratou Rabo Dr Koala Moustapha	Director General Director of Evaluation and Control Director of Regulation, Inspection, and Litigation, ANB Director of the Laboratory of Biosafety/ANB

Category	Institution / Organization	Location	Key Resource Persons	Designation
Government & Regulators	National Seed Committee (NSC)	Ouagadougou	Mr. Prosper Zemba	President
Government & Regulators	National Seed Committee (NSC)	Ouagadougou	Ms. Kadi Zidouemba	Director of crops seeds
Private Sector	Bayer	Ouagadougou	Dr. Karim Traore	Regulatory Manager, Seeds and Trade Africa
Private Sector	Fagri	Ouagadougou	Mr. Issaka Kolga	Director General
Private Sector	EPAM	Ouagadougou	Mr. Raymond Noba	Director General
Private Sector	National Union of Seed Producers of Burkina	Ouagadougou	Mr. Inoussa Ouedraogo	President
Private Sector	NAFASO	Bobo-Dioulasso	Mr. Abdoulaye Sawadogo	President Director General
Private Sector	SEMAFORT		Mr. Bruno Tinland	Director General
NGOs & Advocacy Groups	OFAB Burkina Chapter	Ouagadougou	Dr Edgar Valentin Traore	Former Focal Point
NGOs & Professional Associations	African Society of Genetic	Ouagadougou	Prof. Mahamadou Sawadogo	General Secretary
NGOs & Advocacy Groups	Journalist	Ouagadougou	Mr. Jeremie Ouedraogo	Journalist
NGOs & Advocacy Groups	Organization for Research and Development in Agricultural Science, Technology, and Innovation (RD-STIA)	Ouagadougou	Dr. Valentin Edgar Traoré	Director General
Funders	Africa Agricultural Technology Foundation (AATF)	Abuja (Regional Office)	Dr. Jean Baptiste Tignegre	Regional Director, W/Africa
Funders	National Fund for Research and Innovation for Development (FONRID)	Ouagadougou	Dr. Babou Bationo	Director General